

Motivation

In 1980, Vera Rubin published the first substantial evidence for dark matter via galactic rotation curves. Yet no dark-matter particle has been detected. A second, less-discussed mystery is the **lopsidedness of galaxy spectra**: the kinematic center does not overlap with the luminosity center in more than 35% of observed galaxies.

These two anomalies motivate exploring whether an alternative mechanism — a spatially varying particle mass — can reproduce the observed dynamics without invoking unseen matter.

Hypothesis

All particles share a universal spatial mass profile:

$$m = f \phi(\mathbf{x}),$$

where f is particle-species-specific and $\phi(\mathbf{x})$ is a universal function of position. Near Earth, $\phi \approx \text{const}$, so local experiments cannot falsify the hypothesis. On galactic scales, deviations in ϕ become observable.

Plasma Analogy

The inspiration comes from plasma physics. Photons in a plasma satisfy

$$\omega^2 = \omega_p^2 + k^2,$$

which mirrors Einstein's energy-momentum relation $E^2 = m^2 + p^2$. Identifying the photon mass with ω_p — which depends on local plasma density — shows that a spatially varying density acts as a spatially varying mass. Density gradients cause photon trajectories to curve, interpretable as a mass-gradient effect.

We generalise: if photons acquire an effective space-dependent mass via the medium, other particles (protons, electrons) may be subject to an analogous mechanism on cosmic scales.

Key Insight: A space-dependent mass can produce trajectory deviations that mimic the gravitational effects attributed to dark matter, without requiring new particles.

Methods

One-Dimensional Case

To build the method, we start from a modified relativistic Lagrangian with a variable mass:

$$\mathcal{L} = -m(z)\sqrt{1 - \dot{z}^2}.$$

This is paired with the constant-proper-acceleration trajectory

$$z(t) = \frac{1}{a} \left(\sqrt{a^2 t^2 + 1} - 1 \right).$$

Using the Euler-Lagrange equations and the known trajectory, we invert the problem to recover $m(z)$. The method is validated because it exactly recovers the original (constant) mass. A Lipschitz condition guarantees **uniqueness** at every point in space and time, with an expected singularity at the Rindler horizon.

Two-Dimensional Orbital Case

For orbital mechanics, the modified central-force Lagrangian is

$$\mathcal{L} = \frac{\mu(r) \dot{r}^2}{2} + \frac{\mu(r) r^2 \dot{\phi}^2}{2} + \frac{GmM}{r},$$

where $\mu(r)$ is the position-dependent reduced mass. The Euler-Lagrange equations reduce to an ODE for $\mu(r)$ given an observed trajectory. Analytic inversion is generally intractable, so we compare candidate mass profiles by integrating the equations of motion with a **4th-order Runge-Kutta** scheme.

The effective potentials for the three cases take the form:

$$\begin{aligned} V_{\text{eff, classical}} &= -\frac{GM}{r} + \frac{L^2}{2\mu r^2} \\ V_{\text{eff, relativistic}} &= -\frac{GM}{r} + \frac{L^2}{2\mu r^2} - \frac{GML^2}{\mu r^3} \\ V_{\text{eff, spatial}} &= -\frac{GM}{r} + \frac{L^2}{2\mu(r) r^2} - \frac{GML^2}{\mu(r) r^3} \end{aligned}$$

The RK4 integrator has been tested against known newtonian and relativistic analytic solutions.

Results & Progress

- **1D case complete.** Recovered the trajectory; uniqueness proved via
- **2D non-relativistic potential** mass case and implemented RK4 orbits.
- Analytic two-dimensional solution Schwarzschild action.

Timeline

Period	Milestones
Nov – Feb	Reproduced 1D case; uniqueness
Mar – Apr	2D non-relativistic modified
May	Mass distribution form +
Jun – Jul	Analytic or numerical for
Aug	Begin thesis writing

References

1. Rubin, V. C. *et al.* "Rotational Properties of 21 S
2. Jog, C. J. & Combes, F. "Lopsided Spiral Galaxi
3. Shi, Y. "Force from a Metric with a Space-Depe
4. Hackmann, E. *et al.* "Complete Analytic Solution 171101 (2008).

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